

# Class 16 Dihedral Groups

## Goals

1. Understand structure of dihedral groups.
2. Be able to determine the order of a group and the order of an element in a group.
3. Be able to determine the center of a group and the centralizer of an element.

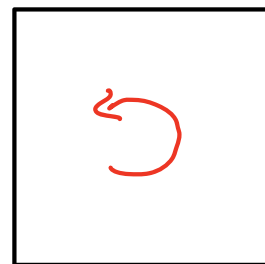
Definition: The dihedral group,  $D_n$ , is the set of all symmetries on a regular  $n$ -sided polygon.

Example:

$D_4$  is the set of symmetries on a square. There are 4 rotation symmetries and 4 reflection symmetries.

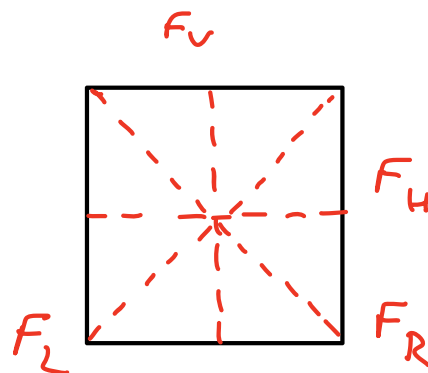
Counterclockwise rotations

$R_0, R_{90}, R_{180}, R_{270}$



Reflections

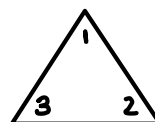
$F_H, F_V, F_R, F_L$



## Dihedral Group Class Activity

Students break up into pairs or triples.

Each group is given an equilateral triangle. Label corners 1, 2, 3 in clockwise order with 1 at the top, 2 on the right and 3 on the left



List all symmetries of the triangle.

| Symmetries of Triangle – Dihedral Group $D_3$ | Permutations of $\{1,2,3\}$ – Symmetric Group<br>$S_3$ |
|-----------------------------------------------|--------------------------------------------------------|
| $R_0$                                         | $(1)$                                                  |
| $R_{120}$                                     | $(1\ 3\ 2)$                                            |
| $R_{240}$                                     | $(1\ 2\ 3)$                                            |
| $F_1$                                         | $(2\ 3)$                                               |
| $F_2$                                         | $(1\ 3)$                                               |
| $F_3$                                         | $(1\ 2)$                                               |

Cayley Table for  $D_3$

|           | $R_0$ | $R_{120}$ | $R_{240}$ | $F_1$ | $F_2$ | $F_3$ |
|-----------|-------|-----------|-----------|-------|-------|-------|
| $R_0$     |       |           |           |       |       |       |
| $R_{120}$ |       |           |           |       |       |       |
| $R_{240}$ |       |           |           |       |       |       |
| $F_1$     |       |           |           |       |       |       |
| $F_2$     |       |           |           |       |       |       |
| $F_3$     |       |           |           |       |       |       |

Cayley Table for  $S_3$

The order of a group,  $G$ , is the number of elements in the group. This is denoted by  $|G|$ .

Examples

$$|D_3| =$$

$$|D_4| =$$

$$|D_n| =$$

$$|\mathbf{Z}_n| =$$

$$|U(10)| =$$

$$|S_n| =$$

The order of an element,  $g$ , is the smallest positive integer  $k$  such that

$$g^k = e \quad \text{Denoted by } |g| = k.$$

Examples

$$\text{In } S_5, |(12)(345)| =$$

$$\text{In } D_4, |R_{270}| =$$

$$\text{In } \mathbf{Z}_{10}, |6| =$$

$$\text{In } U(10), |9| =$$

